

Agentic AI & Multiagent Systems



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Prof Giovanni Merlino
Associate Professor (Computer Engineering)
University Of Messina, Italy

Prof Dr. Om Prakash Vyas
Vice Chancellor & Director
IIIT Naya Raipur, India

Multi-Agent Coordination (MAC)

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Contents

Multi Agent
Coordination

Core Challenges

Coordination
Architectures

Communication
Protocols

Common Multi
Agent Patterns

Human in the
Loop (HITL)

Real World
Applications

Future of
Agentic AI

Multi-Agent Systems

Multi-Agent Systems (MAS) offer a compelling paradigm shift in how we approach complex problem-solving with AI. By distributing tasks across multiple specialized agents, MAS addresses limitations inherent in monolithic, single-agent architectures.

Multi-Agent System Characteristics



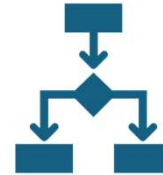
Why Multi-Agent Systems (MAS)?



Divide and Conquer:
Complex tasks split into
sub-tasks.



Context Management:
Distributed memory load
across agents.



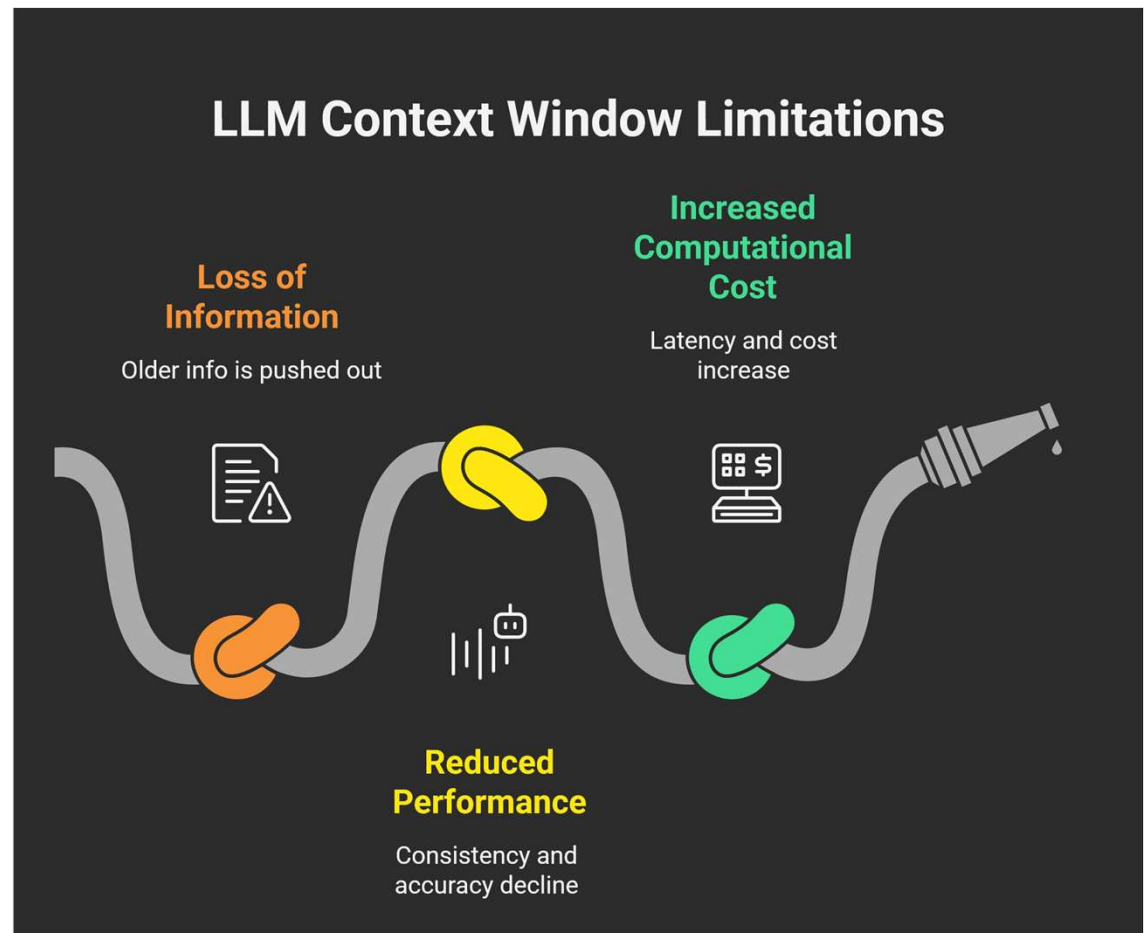
Specialization: Pairing
creative models with
logic-heavy models.



Error Isolation: Critics
can catch hallucinations
before final output.

Context Management

One of the most pressing challenges in working with large language models (LLMs) is the limited context window. Context window refers to the amount of text an LLM can consider when generating a response. As a single agent attempts to handle increasingly complex tasks, it quickly fills its context window with relevant information, past interactions, and intermediate results.



Agent Responsibilities



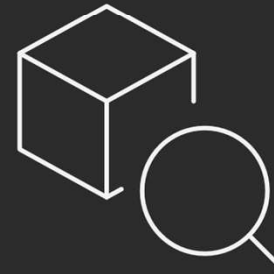
Conversation History

Agent maintains conversation history and identifies key topics.



Customer Profile

Agent retrieves and manages customer information.



Product Knowledge

Agent provides information about specific products.



Policy Enforcement

Agent ensures responses comply with company policies.

Swiss Army Knife Fallacy

SEMANTIC NOISE: LOSS OF NUANCE IN AGENT COMMUNICATION.



INFINITE LOOPS: PREVENTING AGENTS FROM CORRECTING EACH OTHER INDEFINITELY.



STATE MANAGEMENT: MAINTAINING A 'GLOBAL WORKSPACE' FOR STATUS UPDATES.



PROTOCOL STANDARDIZATION: MOVING FROM LOOSE TEXT TO STRUCTURED JSON.

Error Isolation

Hallucination, or the generation of false or misleading information, is a common problem in LLMs. When a single agent hallucinates, it can be difficult to detect and correct the error before the final output is delivered.

MAS provides a mechanism for error isolation by incorporating a "Critic" agent. This agent is responsible for reviewing the output of other agents and identifying any inconsistencies or inaccuracies. If the Critic agent detects a hallucination, it can flag the error and prevent it from propagating to the final output.

Example: Imagine an agent tasked with summarizing a news article. The agent might hallucinate and include information that is not actually present in the article.

Summary Agent: Responsible for generating a summary of the news article.

Critic Agent: Responsible for reviewing the summary and identifying any inaccuracies.

If the Summary Agent hallucinates, the Critic Agent can detect the error and prevent it from being included in the final summary.

Divide and Conquer

Complex tasks can be overwhelming for a single agent to handle. MAS allows you to break down these tasks into smaller, more manageable sub-tasks that can be assigned to individual agents. This "divide and conquer" approach simplifies the overall task and makes it easier to manage and control.

Example: Consider a task that involves planning a complex project, such as building a new website.



The Core Challenges of Coordination



Semantic Noise: Loss of nuance in agent communication.



Infinite Loops: Preventing agents from correcting each other indefinitely.



State Management: Maintaining a 'Global Workspace' for status updates.



Protocol Standardization: Moving from loose text to structured JSON.

Coordination Architectures



Centralized (Manager-Worker): Manager assigns tasks and compiles results.



Decentralized (Peer-to-Peer): Direct negotiation between agents.



Blackboard: Shared memory where agents post and react to data.



Dynamic Teams: Creating/destroying agents based on task complexity.

Manager Agent Features

Single Point of Contact

The user interacts only with the Manager agent, simplifying communication and providing a unified interface.

Centralized Control

The Manager agent holds ultimate authority and responsibility for task allocation, scheduling, and monitoring.



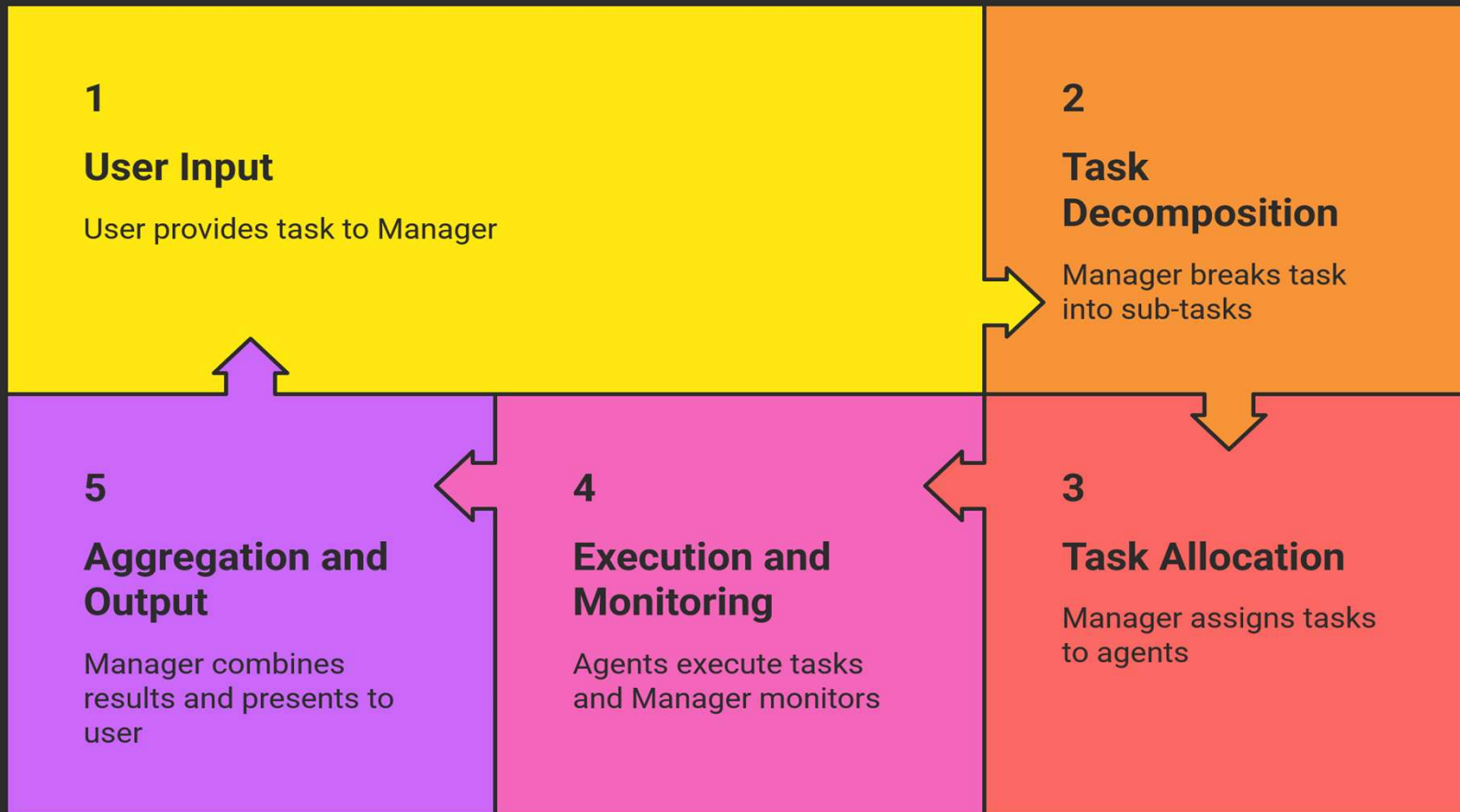
DAG Representation

The relationships between sub-tasks are explicitly defined using a DAG, ensuring proper execution order and dependency management.

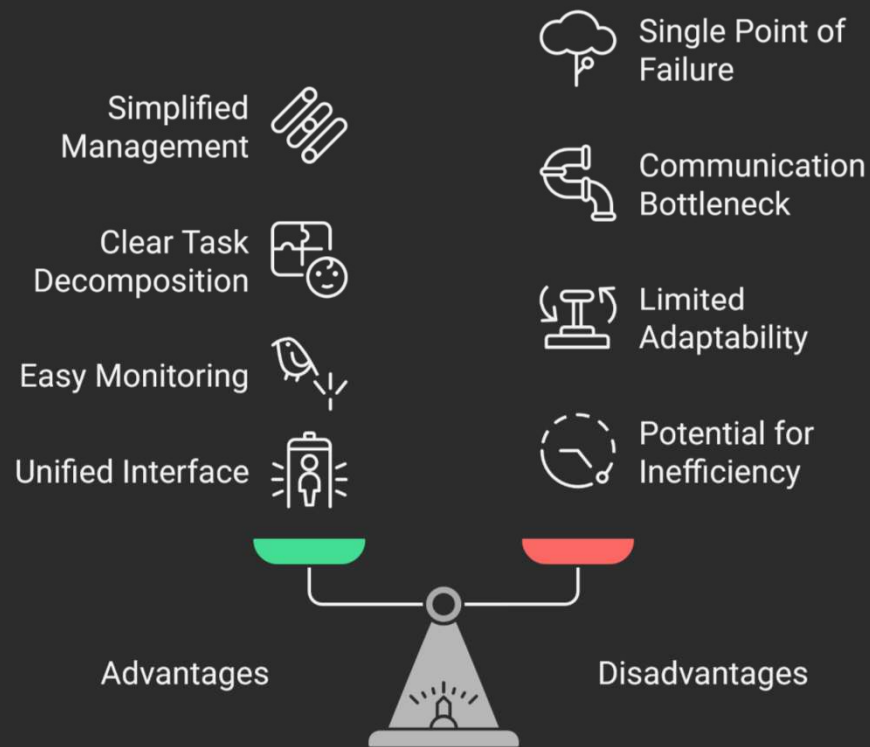
Task Decomposition

Complex tasks are broken down into smaller, more manageable sub-tasks.

Hierarchical Coordination Cycle



Balancing Hierarchical Coordination's Strengths and Weaknesses



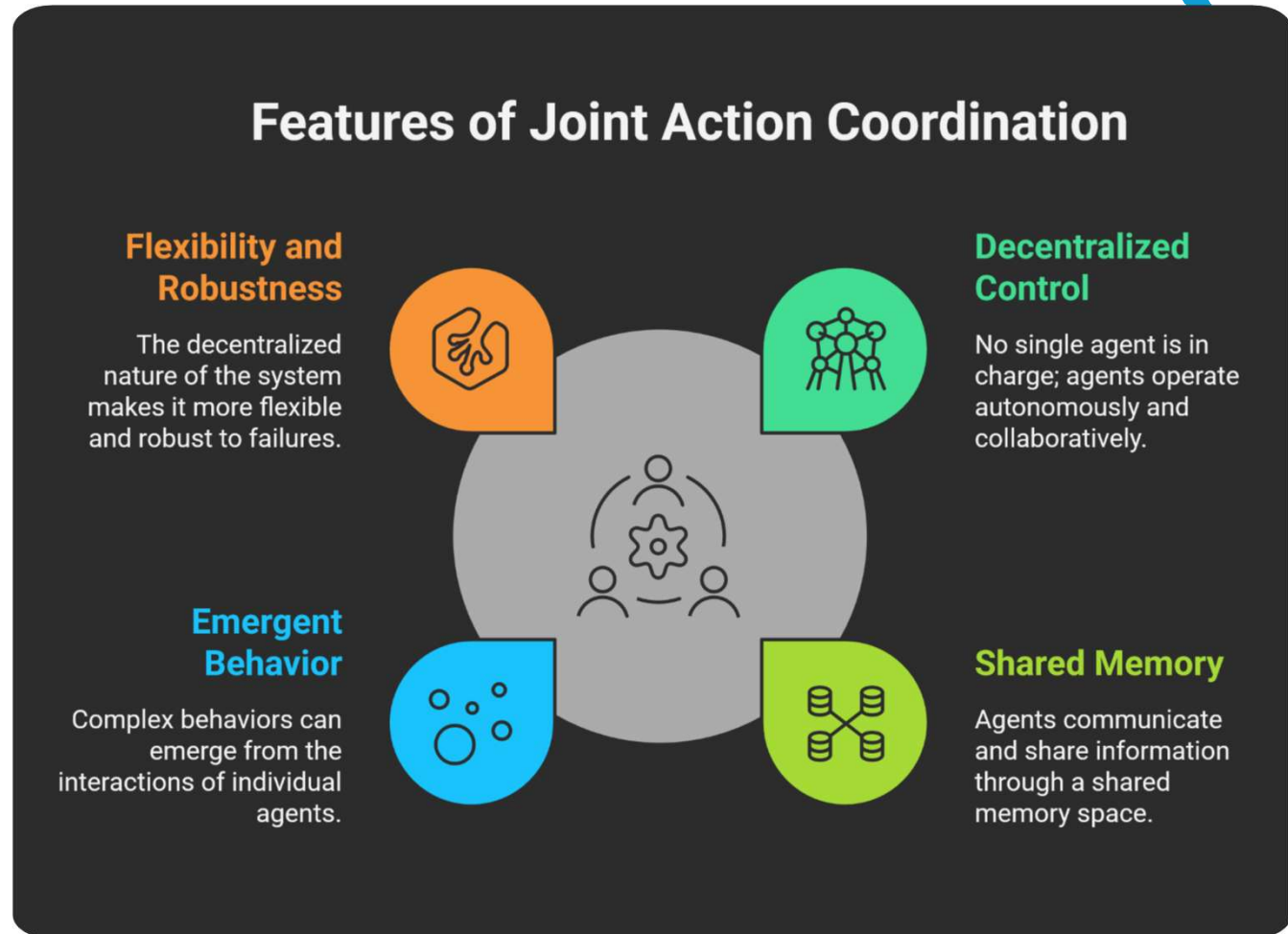
Example:

• Consider a task of planning a trip. The Manager agent might decompose this into sub-tasks such as:

- Researching destinations
- Finding flights
- Booking accommodation
- Creating an itinerary
- Each sub-task would be assigned to a specialized agent (e.g., a "Destination Research Agent," a "Flight Booking Agent," etc.), and the DAG would define the order in which these agents need to execute their tasks.

Joint Action (Peer-to-Peer) Condition

The joint action, or peer-to-peer, coordination architecture relies on a shared memory space, such as a vector database, where agents can contribute and access information. This approach fosters collaboration and emergent behavior, as agents can directly interact and influence each other's actions.



Joint Action Coordination Cycle

Initial State

Agents start with a shared understanding of the goal.

Iteration

Agents repeat the process to refine understanding.

Execution and Observation

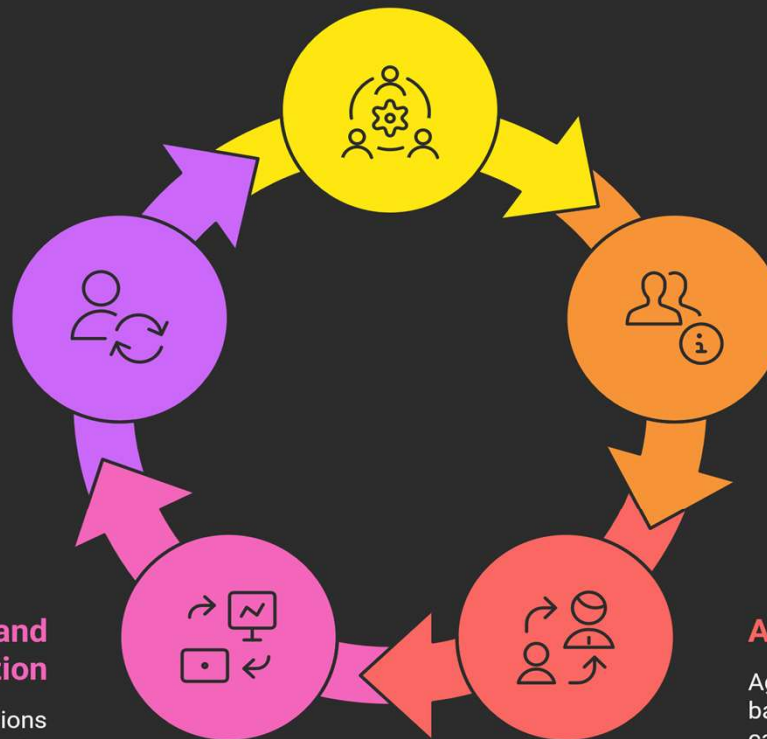
Agents perform actions and observe effects.

Information Sharing

Agents contribute knowledge to shared memory.

Action Selection

Agents choose actions based on goals and capabilities.



Dynamic Teams Coordination

Dynamic teams represent a more modern and flexible approach to multi-agent coordination. Systems like AutoGen allow agents to be created or destroyed on the fly based on the complexity of the task. This dynamic allocation of resources enables the system to adapt to changing requirements and optimize performance.

Dynamic Team Coordination Overview

Improved Efficiency

System achieves higher efficiency by adapting to task requirements

Adaptive Agent Creation

Agents are created or destroyed based on task complexity

Resource Optimization

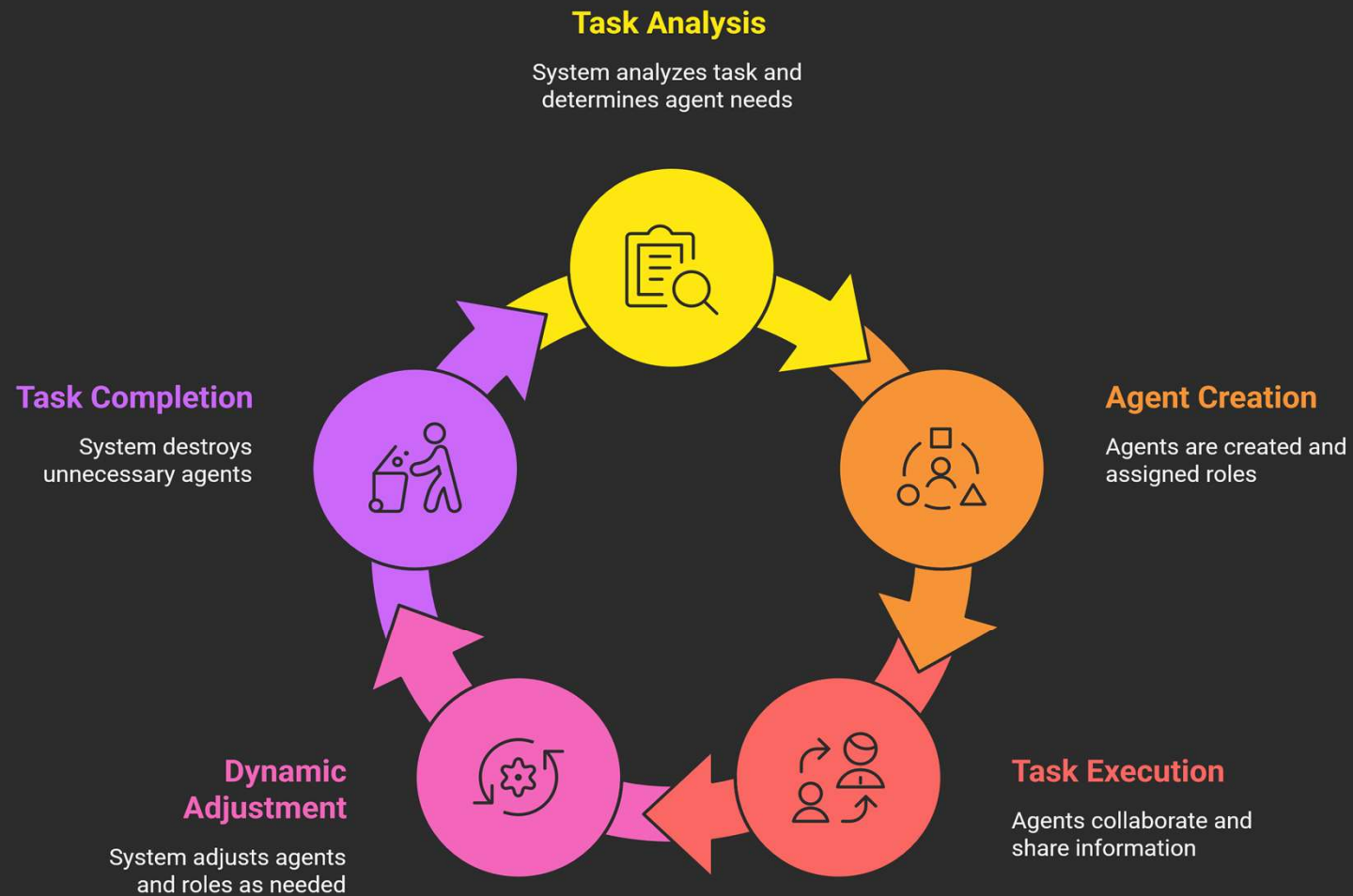
System optimizes resource allocation by adjusting agent numbers

Dynamic Role Assignment

Agents can be assigned different roles during task execution



Dynamic Team Coordination Cycle



Communication Protocols



Syntax vs. Semantics:
Format (JSON) vs.
Meaning.



Pub/Sub Models: Agents
subscribe to specific data
categories.



Asynchronous Workflows:
Agents working in parallel
to save time.



Token Efficiency:
Optimizing dialogue to
reduce cost and latency.

Common Multi-Agent Patterns

Debate Pattern: Two agents argue to reach a logical conclusion.

Actor-Critic: One generates, one critiques, first one improves.

Sandwich Pattern: Simple agents handle prep/cleanup; complex agents handle core.

Orchestration: A lead agent creates a Directed Acyclic Graph (DAG) for tasks.

Human-in-the-Loop (HITL)



Human-as-an-Agent: The human is the 'Ultimate Authority' agent.

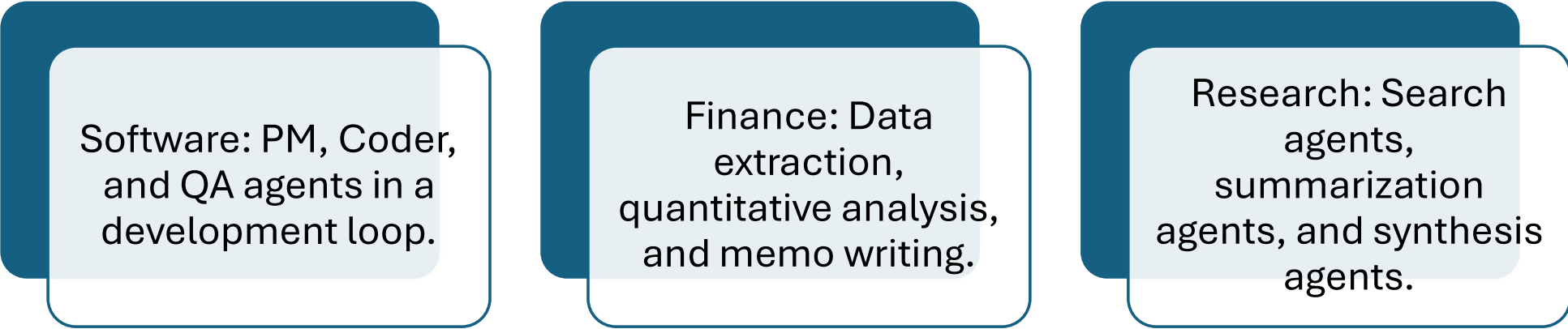


Intervention Gates: Approvals needed for high-stakes actions.



Alignment: Humans steer the 'society of agents' toward ethical outcomes.

Real-World Applications

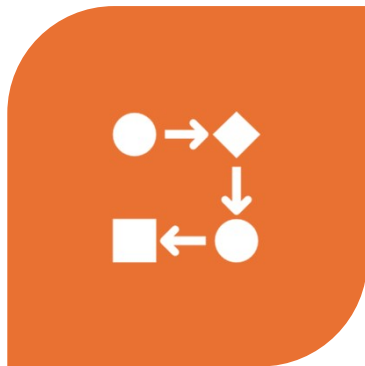


Software: PM, Coder, and QA agents in a development loop.

Finance: Data extraction, quantitative analysis, and memo writing.

Research: Search agents, summarization agents, and synthesis agents.

The Future of Agentic AI



BEYOND THE CHATBOT: SHIFTING
FROM PROMPTS TO
WORKFLOWS.



SCALING INTELLIGENCE:
ACHIEVING RESULTS GREATER
THAN ANY SINGLE MODEL.



TAKEAWAY: SUCCESS =
STRUCTURED COMMUNICATION
+ DIVERSE ROLES.